

XALQARO PUL O'TKAZMALARI BIRLASHTIRILGAN MONITORING VA SVERKA TIZIMI

Bank bek-ofisi uchun texnik hujjat (TZ)

Tashkilot turi: kommertsiya banki

Asosiy foydalanuvchi: bek-ofis / sverka xodimi

Backend: .NET 10 | Frontend: Vue 3 | DB: PostgreSQL

Sana: 12.06.2026

Mundarija

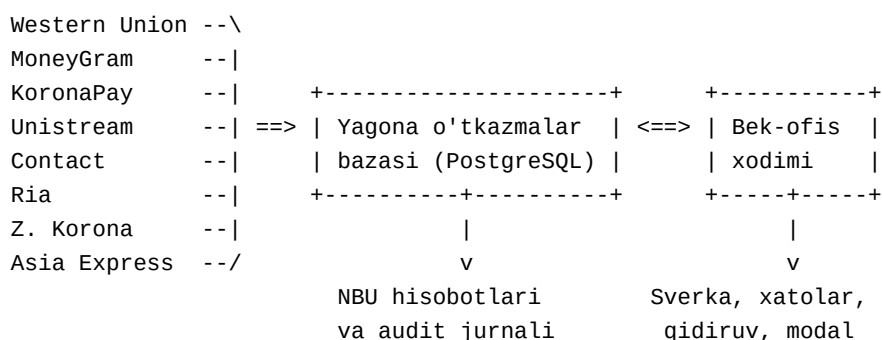
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1. Loyihaning maqsadi

Bank bek-ofisi har kuni 8 ta xalqaro pul o'tkazma tizimi bilan ishlaydi: Western Union, MoneyGram, KoronaPay, Unistream, Contact, Ria, Zolotaya Korona va Asia Express. Har bir tizimning o'z portal, o'z hisobot fayli va o'z formati bor. Bu vaziyatda kunlik sverka, xatolarni tahlil qilish va NBU hisobotini tayyorlash o'ta mehnattalab.

Loyihaning maqsadi - barcha partnyor tizimlardan kelgan o'tkazmalarni yagona ma'lumotlar bazasida to'plash va bek-ofis xodimiga bitta veb interfeys orqali ko'rish, qidirish, sverka qilish va hisobot olish imkonini beradigan ichki bank tizimini yaratish.

Asosiy g'oya



2. Hal qilinadigan muammolar

Hujjatdagi har bir funksiya quyidagi 3 ta asosiy muammodan bittasini yechadi. Ro'yxat aniq:

Muammo 1. Sverka (reconciliation) qo'lda qilinmoqda

Hozirgi holat:

- Har kuni har bir partnyordan Excel/CSV fayl yuklanadi.
- Xodim qo'lda ABS bilan solishtiradi: PaymentRef, summa, sana.
- 1 partnyor uchun ~200-500 ta o'tkazma, 8 partnyor x 1-2 soat.
- Inson xatosi yuqori, farqlarni topish kech bo'ladi.

Yechim (tizim):

- Partnyor faylini yuklab, avtomatik moslash (matching).
- 'Faqat partnyorda bor', 'faqat bizda bor', 'summa farq qiladi' kategoriyalari avtomatik ajraladi.
- Sverka aktiga 1 tugma bilan tayyor PDF/Excel chiqaradi.

Muammo 2. Xato o'tkazmalarni kuzatish qiyin

Hozirgi holat:

- Xato bo'lgan o'tkazmalar har bir partnyor portalida alohida.
- Xato sabablari turli formatda (kod, matn, til).
- Mijozga javob berish uchun xodim har bir portalga qaraydi.

Yechim (tizim):

- Barcha xato o'tkazmalar yagona dashboardda.
- Xato sabablari me'yorlashtirilgan toifaga ajratiladi:
KYC_FAIL, AML_HOLD, BENEFICIARY_NOT_FOUND, LIMIT_EXCEEDED, TECH_TIMEOUT, CANCELLED_BY_SENDER va h.k.
- Statistika: qaysi partnyorda nechta, qaysi sababdan ko'p.

Muammo 3. Audit / NBU hisoboti uchun tarix yo'q

Hozirgi holat:

- Tarix har bir partnyor portalida 30-90 kun saqlanadi.
- Eski o'tkazmalarni topish uchun partnyorga so'rov yuboriladi.
- NBU forma 402 va shubhali operatsiyalar (AML) hisoboti uchun ma'lumot yig'ish kunlab vaqt oladi.

Yechim (tizim):

- O'tkazmalar bizning bazada 7 yil saqlanadi (NBU talabi).
- Har qanday o'zgarish (status, sverka holati) audit jurnalida yoziladi: kim, qachon, nima qildi.
- NBU hisoboti tugma orqali tayyor formatda chiqariladi.

3. Tizim foydalanuvchilari va rollari

Ro'l	Kim	Asosiy harakatlar
sverka_officer	Bek-ofis sverka xodimi (asosiy aktor)	Ro'yxat, modal, sverka, partnyor fayl yuklash, farqlarni hal qilish.
compliance	Compliance/AML xodimi	Xato/shubhali o'tkazma tahlili, AML hisoboti.
auditor	Ichki audit / NBU bilan bog'lovchi	Faqat o'qish; tarix, jurnal, hisobot eksport.
admin	IT administrator	Foydalanuvchi, partnyor va tizim sozlamalari.

Eslatma: Mijozga to'g'ridan-to'g'ri ko'rinish bermaymiz - bu ichki bank tizimi.

4. Foydalanuvchi senariylari (use-case)

UC-1. Kunlik sverka

Aktor: sverka_officer

Boshlanish: ish kuni boshida (08:30).

Qadamlar:

1. Tizimga kiradi.
2. 'Sverka' bo'limini ochadi, partnyorni tanlaydi (mas. KoronaPay).
3. Partnyordan kelgan kunlik faylni (CSV/Excel) yuklaydi.
4. Tizim avtomatik moslashtiradi va 4 ta toifaga ajratadi:
 - moslashgan (matched)
 - faqat partnyorda (missing_in_bank)
 - faqat bankda (missing_in_partner)

- summa/sana farq qiladi (mismatch)

5. Har bir nomoslash uchun izoh yozadi yoki 'tekshiruvga' belgilaydi.

6. Sverka aktini PDF qilib chiqaradi va imzolaydi.

Natija: kunlik sverka 1-2 soat o'rniga 10-15 daqiqada bajariladi.

UC-2. Xato o'tkazma sababini topish

Aktor: compliance / sverka_officer

Boshlanish: mijoz qo'ng'iroq qildi yoki dashboardda xato ko'rindi.

Qadamlar:

1. Asosiy ekranda 'Xatolar' filtrini yoqadi.
2. PaymentRef yoki MTCN bo'yicha qidiradi.
3. Modal ochadi - xato kodi, sababi, partnyor javobi ko'rinadi.
4. Agar AML_HOLD bo'lsa, compliance bo'limiga yuboradi.
5. Agar texnik bo'lsa, qayta yuborish (retry) tugmasi.

Natija: xatoning sababi 1 ekranda, partnyor portaliga kirish shart emas.

UC-3. NBU hisoboti tayyorlash

Aktor: auditor

Boshlanish: oy oxirida, NBU forma 402 talab qilinadi.

Qadamlar:

1. 'Hisobotlar' bo'limiga kiradi.
2. 'NBU 402' shablonini tanlaydi.
3. Sana oralig'ini, valyutani belgilaydi.
4. Tizim jamlagan ma'lumotni Excel formatida chiqaradi.

Natija: hisobot 2-3 kun o'rniga 5 daqiqada tayyor.

5. Funktsional talablar

Quyidagi har bir funksiya qaysi muammoni yechishi izohi bilan berilgan (M1=sverka, M2=xato tahlili, M3=audit/NBU).

#	Funksiya	Yechim	Asosiy aktor
F1	Yagona o'tkazmalar ro'yxati (jadval)	M1, M2	sverka_officer
F2	Filtr: partnyor, status, sana, valuta	M1, M2	sverka_officer
F3	Qidiruv: PaymentRef / MTCN / passport	M2	compliance
F4	Modal: o'tkazma to'liq tafsiloti	M2	sverka_officer
F5	Partnyor fayl yuklash (CSV/Excel)	M1	sverka_officer
F6	Avtomatik moslash + farqlar ro'yxati	M1	sverka_officer
F7	Sverka aktini PDF qilib chiqarish	M1, M3	sverka_officer
F8	Xato statistikasi (sabab/partnyor)	M2	compliance
F9	Audit jurnali (ko'rish, eksport)	M3	auditor
F10	NBU forma 402 hisoboti (Excel)	M3	auditor
F11	Foydalanuvchi va partnyor boshqaruvi	-	admin

6. Xalqaro o'tkazmaning ma'lumot maydonlari

Xalqaro pul o'tkazmasi mahalliy o'tkazmadan farq qiladi: korridor, valyuta jufti, MTCN/Reference, o'tkazma yo'nalishi (kiruvchi yoki chiquvchi), tomon banki/agenti kabi maydonlar bor.

Maydon	Tavsif	
payment_ref	Bizning ichki ID (UUID)	
partner_code	WU, MGR, KP, UNI, CON, RIA, ZK, ASE	
partner_txn_id	Partnyordagi original ID	
mtcn	Money Transfer Control Number (WU/MGR)	
direction	INCOMING / OUTGOING	
status	created, sent, pending, paid, cancelled,	
	failed, refunded	
corridor_from / _to	Davlat ISO kodlari (UZ, RU, US, TR, ...)	
send_currency / amount	Jo'natilgan valyuta va summa	
recv_currency / amount	Qabul valyutasi va summa	
amount_uzs	UZSga o'girilgan ekvivalent	
fx_rate	Qo'llanilgan kurs	
usd_rate	NBU/CBU dollar kursi	
commission	Komissiya (jo'natuvchi tomonidan)	
sender_*	F.I.Sh, passport seriya/raqam, telefon,	
	tug'ilgan sana, fuqarolik, manzil	
receiver_*	F.I.Sh, passport, telefon, fuqarolik	
sender_country	Jo'natuvchi davlati	
receiver_country	Qabul qiluvchi davlati	
purpose_code	O'tkazma maqsadi (oilaviy, savdo, ...)	
sanctions_check	clean / flagged / blocked	
aml_score	0-100	
error_code / message	Xato kodi va matni (me'yorlashgan)	
sent_at / paid_at	Sanalar	
recon_status	not_checked, matched, mismatch, missing	
recon_batch_id	Sverka partiyasi ID	

7. Sverka (reconciliation) oqimi

[1. Partnyor fayli (CSV/Excel)]

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[2. Parser - partnyor formatiga maxsus]

- WU adapteri - Unistream adapteri
- MoneyGram adapteri - Contact adapteri
- KoronaPay adapteri - Ria, Z. Korona, Asia Express

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[3. Matching engine]

kalit: partner_txn_id YOKI mtcn

to'lqinlar:

- a) bir xil ID + summa + sana -> matched
- b) bir xil ID, summa farqi -> amount_mismatch
- c) faqat partnyorda -> missing_in_bank
- d) faqat bankda -> missing_in_partner

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[4. Sverka partiyasi (recon_batch)]

- sana: 11.06.2026
- partnyor: KoronaPay
- jami: 412 ta yozuv

```

- matched: 405, mismatch: 3, missing_in_bank: 2,
  missing_in_partner: 2
  |
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[5. Xodim har bir farqga izoh yozadi yoki tekshiruvga yuboradi]
  |
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[6. Sverka akti (PDF) - imzo va arxiv]

```

8. Xato o'tkazmalar bilan ishlash

Har bir partnyorning o'z xato kodi bo'ladi. Tizim ularni ichki, me'yorlashgan toifalarga moslab oladi - xodim faqat toifa bo'yicha ishlasa bo'ladi.

Ichki toifa	Tavsif va misol
KYC_FAIL	Mijoz hujjati notog'ri / muddati o'tgan
AML_HOLD	AML/sanksiya ro'yxati - compliance ko'radi
BENEFICIARY_NOT_FOUND	Qabul qiluvchi topilmadi yoki MTCN yaroqsiz
LIMIT_EXCEEDED	Kunlik/oylik limit oshib ketdi
INSUFFICIENT_FUNDS	Mablag' yetmadi (chiquvchida)
TECH_TIMEOUT	Texnik xato, partnyor javobi yo'q
CANCELLED_BY_SENDER	Jo'natuvchi bekor qildi
FX_RATE_MISMATCH	Kurs farqi limitdan oshdi
OTHER	Boshqa (matn ichida saqlanadi)

Statistika ko'rinishi

- Partnyor x toifa kesimida xato soni (oxirgi 30 kun).
- Trend grafigi - kun bo'yicha xato foizi.
- Top 10 takrorlanuvchi xato matni (asl, partnyor matni).

9. Audit jurnali va NBU hisoboti

9.1. Audit jurnali (audit_log jadvali)

Tizimdagi har qanday muhim harakat yoziladi. Hech qanday yozuv o'chirilmaydi yoki o'zgartirilmaydi - faqat yangi yozuv qo'shiladi.

```

audit_log:
- id, timestamp, user_id, user_login
- action: LOGIN, VIEW_PAYMENT, EXPORT_REPORT, RUN_RECON,
  RESOLVE_DIFF, RETRY_PAYMENT, CHANGE_USER, ...
- entity_type, entity_id (qaysi obyekt ustida)
- details_json (eski va yangi qiymat)
- ip_address

```

9.2. NBU hisobotlari (chiqariladigan formatlar)

- Forma 402 - xalqaro o'tkazmalar bo'yicha kunlik hisobot (Excel).
- AML hisoboti - flagged va blocked o'tkazmalar (PDF + Excel).
- Korridor kesimida statistik hisobot (kelgan/ketgan, davlat).
- Saqlash muddati: 7 yil (NBU talabi); arxivga ko'chiriladi.

10. Tizim arxitekturası

Partnyor ma'lumot manbalari

=====

- Avtomatik: API/Webhook (qaerda mavjud bo'lsa)
- Yarim avtomatik: SFTP/elektron pochta orqali kunlik fayl
- Qo'lda: xodim tomonidan UI orqali yuklash

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Integration / Import qatlami	(.NET 10 background services)
- WU adapter	
- MoneyGram adapter	
- KoronaPay adapter	
- Unistream / Contact / Ria /	
Z. Korona / Asia Express	

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| yagona ichki model (Payment)

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Domain qatlami	<-----> PostgreSQL (asosiy DB)	
- Reconciliation engine		payments, partners,
- Error normalizer		recon_batches,
- Audit logger		recon_diffs, audit_log
- Report generator (NBU)		

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API qatlami (.NET 10 Web API)	JWT, RBAC
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Vue 3 SPA (bek-ofis xodimi)	jadval, modal, sverka, hisobot
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11. Ma'lumotlar bazasi (SQL)

Asosiy jadvallar (PostgreSQL). Maxfiy maydonlar shifrlanadi (PII).

```
CREATE TABLE roles (  
    id SERIAL PRIMARY KEY,  
    name VARCHAR(40) UNIQUE NOT NULL  
    -- 'sverka_officer','compliance','auditor','admin'  
);  
  
CREATE TABLE partners (  
    id SERIAL PRIMARY KEY,  
    code VARCHAR(10) UNIQUE NOT NULL,    -- WU, MGR, KP, UNI, CON, RIA, ZK, ASE  
    name VARCHAR(100) NOT NULL,  
    is_active BOOLEAN DEFAULT true  
);  
  
CREATE TABLE users (  
    id SERIAL PRIMARY KEY,  
    name VARCHAR(40) UNIQUE NOT NULL,  
    email VARCHAR(100) UNIQUE NOT NULL,  
    password VARCHAR(100) NOT NULL,  
    is_active BOOLEAN DEFAULT true  
);
```

```

    id SERIAL PRIMARY KEY,
    login VARCHAR(100) UNIQUE NOT NULL,
    password_hash VARCHAR(255) NOT NULL,
    full_name VARCHAR(150),
    role_id INT NOT NULL REFERENCES roles(id),
    is_active BOOLEAN DEFAULT true,
    created_at TIMESTAMP DEFAULT now()
);

CREATE TABLE payments (
    id BIGSERIAL PRIMARY KEY,
    payment_ref      VARCHAR(64) UNIQUE NOT NULL, -- ichki UUID
    partner_id       INT NOT NULL REFERENCES partners(id),
    partner_txn_id   VARCHAR(100),
    mtcn             VARCHAR(40),                  -- WU/MGR
    direction        VARCHAR(10) NOT NULL,         -- INCOMING/OUTGOING
    status           VARCHAR(20) NOT NULL,
    -- korridor / valyuta
    corridor_from    CHAR(2),                      -- ISO: UZ, RU
    corridor_to      CHAR(2),
    send_currency    CHAR(3), send_amount          NUMERIC(18,2),
    recv_currency    CHAR(3), recv_amount          NUMERIC(18,2),
    amount_uzs       NUMERIC(18,2),
    fx_rate          NUMERIC(14,6),
    usd_rate         NUMERIC(14,4),
    commission       NUMERIC(18,2),
    -- jo'natuvchi (PII shifrlanadi)
    sender_full_name VARCHAR(200),
    sender_passport  VARCHAR(40),
    sender_phone     VARCHAR(20),
    sender_country   CHAR(2),
    -- qabul qiluvchi
    receiver_full_name VARCHAR(200),
    receiver_passport VARCHAR(40),
    receiver_phone    VARCHAR(20),
    receiver_country  CHAR(2),
    -- compliance
    purpose_code      VARCHAR(20),
    sanctions_check   VARCHAR(20) DEFAULT 'clean',
    aml_score         SMALLINT,
    -- xato
    error_category    VARCHAR(40), -- KYC_FAIL, AML_HOLD ...
    error_code        VARCHAR(50),
    error_message     TEXT,
    -- sverka
    recon_status      VARCHAR(20) DEFAULT 'not_checked',
    recon_batch_id    BIGINT,
    -- vaqt
    sent_at           TIMESTAMP,
    paid_at           TIMESTAMP,
    created_at        TIMESTAMP DEFAULT now(),
    updated_at        TIMESTAMP DEFAULT now()
);

CREATE INDEX ix_payments_partner ON payments(partner_id);
CREATE INDEX ix_payments_mtcn   ON payments(mtcn);
CREATE INDEX ix_payments_status ON payments(status);

```

```

CREATE INDEX ix_payments_sent_at    ON payments(sent_at);
CREATE INDEX ix_payments_recon      ON payments(recon_status, recon_batch_id);

-- Sverka partiyalari va farqlari
CREATE TABLE recon_batches (
    id BIGSERIAL PRIMARY KEY,
    partner_id INT NOT NULL REFERENCES partners(id),
    business_date DATE NOT NULL,
    file_name VARCHAR(255),
    total_partner INT, total_bank INT,
    matched INT, mismatch INT,
    missing_in_bank INT, missing_in_partner INT,
    status VARCHAR(20) DEFAULT 'open', -- open/closed/signed
    created_by INT REFERENCES users(id),
    created_at TIMESTAMP DEFAULT now()
);

CREATE TABLE recon_diffs (
    id BIGSERIAL PRIMARY KEY,
    batch_id BIGINT NOT NULL REFERENCES recon_batches(id),
    payment_id BIGINT REFERENCES payments(id),
    diff_type VARCHAR(30) NOT NULL,
    -- amount_mismatch / missing_in_bank / missing_in_partner
    partner_record_json JSONB,
    bank_record_json    JSONB,
    resolution VARCHAR(20), -- pending/accepted/disputed
    note TEXT,
    resolved_by INT REFERENCES users(id),
    resolved_at TIMESTAMP
);

CREATE TABLE audit_log (
    id BIGSERIAL PRIMARY KEY,
    ts TIMESTAMP DEFAULT now(),
    user_id INT REFERENCES users(id),
    user_login VARCHAR(100),
    action VARCHAR(40) NOT NULL,
    entity_type VARCHAR(40),
    entity_id VARCHAR(64),
    details JSONB,
    ip_address VARCHAR(45)
);

```

12. Backend modellar (.NET 10)

Loyiha tuzilishi:

```

InternationalTransfers.Api/
|- Domain/
|   |- Entities/    (Payment, Partner, User, Role,
|   |               ReconBatch, ReconDiff, AuditLog)
|   |- Enums/      (Direction, PaymentStatus, ReconStatus,
|   |               ErrorCategory, DiffType)
|   |- Services/   (ReconciliationService, ImportService,
|   |               ErrorNormalizer, ReportService, AuditService)
|- Infrastructure/
|   |- Persistence/ (AppDbContext, EF Core configs)

```

```
|   |- Adapters/           (WuAdapter, MoneyGramAdapter, ...)
|- Application/
|   |- Dtos/, Queries/, Commands/
|- Api/
|   |- Controllers/       (PaymentsController, ReconciliationController,
|                           ReportsController, AuthController)
|   |- Program.cs
|- appsettings.json
```

Asosiy entity: Payment.cs (qisqartma)

```
public class Payment
{
    public long Id { get; set; }
    public string PaymentRef { get; set; } = default!;
    public int PartnerId { get; set; }
    public Partner? Partner { get; set; }
    public string? PartnerTxnId { get; set; }
    public string? Mtcn { get; set; }
    public Direction Direction { get; set; }
    public PaymentStatus Status { get; set; }

    public string? CorridorFrom { get; set; }
    public string? CorridorTo { get; set; }
    public string? SendCurrency { get; set; }
    public decimal? SendAmount { get; set; }
    public string? RecvCurrency { get; set; }
    public decimal? RecvAmount { get; set; }
    public decimal? AmountUzs { get; set; }
    public decimal? FxRate { get; set; }
    public decimal? UsdRate { get; set; }
    public decimal? Commission { get; set; }

    // jo'natuvchi va qabul qiluvchi PII (shifrlanadi)
    public string? SenderFullName { get; set; }
    public string? SenderPassport { get; set; }
    public string? SenderPhone { get; set; }
    public string? SenderCountry { get; set; }
    public string? ReceiverFullName { get; set; }
    public string? ReceiverPassport { get; set; }
    public string? ReceiverPhone { get; set; }
    public string? ReceiverCountry { get; set; }

    public string? PurposeCode { get; set; }
    public string SanctionsCheck { get; set; } = "clean";
    public short? AmlScore { get; set; }

    public ErrorCategory? ErrorCategory { get; set; }
    public string? ErrorCode { get; set; }
    public string? ErrorMessage { get; set; }

    public ReconStatus ReconStatus { get; set; } = ReconStatus.NotChecked;
    public long? ReconBatchId { get; set; }

    public DateTime? SentAt { get; set; }
    public DateTime? PaidAt { get; set; }
    public DateTime CreatedAt { get; set; } = DateTime.UtcNow;
```

```

    public DateTime UpdatedAt { get; set; } = DateTime.UtcNow;
}

```

ReconBatch.cs va ReconDiff.cs

```

public class ReconBatch
{
    public long Id { get; set; }
    public int PartnerId { get; set; }
    public DateOnly BusinessDate { get; set; }
    public string? FileName { get; set; }
    public int TotalPartner { get; set; }
    public int TotalBank { get; set; }
    public int Matched { get; set; }
    public int Mismatch { get; set; }
    public int MissingInBank { get; set; }
    public int MissingInPartner { get; set; }
    public string Status { get; set; } = "open";
    public int? CreatedBy { get; set; }
    public DateTime CreatedAt { get; set; } = DateTime.UtcNow;
    public List<ReconDiff> Diffs { get; set; } = new();
}

public class ReconDiff
{
    public long Id { get; set; }
    public long BatchId { get; set; }
    public long? PaymentId { get; set; }
    public DiffType DiffType { get; set; }
    public string? PartnerRecordJson { get; set; }
    public string? BankRecordJson { get; set; }
    public string Resolution { get; set; } = "pending";
    public string? Note { get; set; }
}

```

Enumlar

```

public enum Direction { Incoming, Outgoing }

public enum PaymentStatus {
    Created, Sent, Pending, Paid, Cancelled, Failed, Refunded
}

public enum ReconStatus {
    NotChecked, Matched, AmountMismatch,
    MissingInBank, MissingInPartner
}

public enum ErrorCategory {
    KYC_FAIL, AML_HOLD, BENEFICIARY_NOT_FOUND, LIMIT_EXCEEDED,
    INSUFFICIENT_FUNDS, TECH_TIMEOUT, CANCELLED_BY_SENDER,
    FX_RATE_MISMATCH, OTHER
}

public enum DiffType {
    AmountMismatch, MissingInBank, MissingInPartner
}

```

13. API endpointlar

Metod	URL	Rol(lar)
POST	/api/auth/login	hammasi
GET	/api/payments	sverka, compliance, auditor
GET	?partner=&status=&from=&to=&q=&recon=&error_cat=&page=&limit=	
GET	/api/payments/{id}	sverka, compliance, auditor
POST	/api/payments/{id}/retry	sverka, compliance
GET	/api/errors/stats	compliance
POST	?from=&to=&partner=	
POST	/api/recon/upload	sverka
	(multipart: file + partner_id)	
GET	/api/recon/batches	sverka, auditor
GET	/api/recon/batches/{id}	sverka, auditor
GET	/api/recon/batches/{id}/diffs	sverka
POST	/api/recon/diffs/{id}/resolve	sverka
POST	/api/recon/batches/{id}/sign	sverka
GET	/api/recon/batches/{id}/act.pdf	sverka, auditor
GET	/api/reports/nbu-402.xlsx	auditor
	?from=&to=	
GET	/api/reports/aml.xlsx	compliance, auditor
GET	/api/audit	auditor
	?user=&action=&from=&to=	

14. Frontend (Vue 3)

SPA, Composition API, Pinia + Vue Router. Asosiy sahifalar:

```
frontend/src/views/
  LoginView.vue
  PaymentsView.vue          (F1, F2, F3 - asosiy ish ekrani)
  PaymentModalView.vue      (F4 - to'liq tafsilot)
  ReconciliationView.vue    (F5, F6, F7 - sverka)
    |- ReconUpload.vue
    |- ReconBatchList.vue
    |- ReconBatchDetail.vue
  ErrorsDashboard.vue        (F8 - xato statistikasi)
  ReportsView.vue           (F10 - NBU 402, AML)
  AuditView.vue             (F9 - audit jurnali)
  AdminView.vue             (F11 - foydalanuvchilar, partnyorlar)
```

15. Xavfsizlik va kirish nazorati

- RBAC - har bir endpoint rol(lar)i bilan himoyalangan; tekshiruv serverda.
- Parol BCrypt hash, JWT amal qilish muddati 8 soat.
- PII (passport, telefon) ma'lumotlar bazasida shifrlanadi (AES-256, kalit Vault yoki AWS KMS'da).
- Modal'da karta/passport maskalangan; to'liq qiymat faqat compliance va sverka

rolida ko'rinadi.

- Audit_log o'chirilmaydi; faqat append. Saqlash 7 yil.
- Sanksiya ro'yxati (OFAC/EU/NBU) bilan har bir o'tkazma tekshiriladi.
- CORS faqat ishonchli ichki domenga ochiq, HTTPS majburiy.

16. Bajarilish ketma-ketligi (rejasi)

Sprint 1 (yadro, 2 hafta):

- .NET 10 va Vue 3 skelet, JWT, RBAC.
- DB migratsiyalari (payments, partners, users, roles, audit_log).
- Login, foydalanuvchilar va partnyorlar boshqaruvi.

Sprint 2 (asosiy ko'rish, 2 hafta): -> M2 ning bir qismini yopadi

- Yagona o'tkazmalar ro'yxati (F1).
- Filtr va qidiruv (F2, F3).
- To'liq tafsilot modali (F4).

Sprint 3 (sverka, 2-3 hafta): -> M1 ni yopadi

- Partnyor adapterlari (kamida 3 ta partnyor uchun parser).
- Matching engine, recon_batches, recon_diffs.
- Fayl yuklash UI, farqlar ekrani, izoh va resolve.
- Sverka akti PDF.

Sprint 4 (xatolar va NBU, 2 hafta): -> M2 va M3 ni yopadi

- Xato me'yorlashtiruvchi, statistika dashboardi (F8).
- NBU 402 va AML hisobotlari (F10).
- Audit jurnali ekrani (F9).

Sprint 5 (qolgan partnyor adapterlari + qattiqlash, 1-2 hafta):

- Qolgan partnyor adapterlari.
- PII shifrlash, sanksiya integratsiyasi, yuklamani sinash.

Umumiy taxminiy muddat: 9-11 hafta.

17. Loyihani ishga tushirish

Backend (.NET 10)

```
dotnet new webapi -n InternationalTransfers.Api
cd InternationalTransfers.Api
dotnet add package Microsoft.EntityFrameworkCore
dotnet add package Npgsql.EntityFrameworkCore.PostgreSQL
dotnet add package Microsoft.AspNetCore.Authentication.JwtBearer
dotnet add package BCrypt.Net-Next
dotnet add package ClosedXML           # Excel hisobot
dotnet add package QuestPDF            # PDF (sverka akti)

dotnet ef migrations add Init
dotnet ef database update
dotnet run
```

Frontend (Vue 3)

```
npm create vite@latest frontend -- --template vue
cd frontend
```

```
npm install axios pinia vue-router element-plus
npm run dev # http://localhost:5173
```

appsettings.json (namuna)

```
{
  "ConnectionStrings": {
    "Default": "Host=localhost;Port=5432;Database=intl_transfers;" +
      "Username=postgres;Password=postgres"
  },
  "Jwt": { "Key": "BU_YERGA_KUCHLI_MAXFIY_KALIT_QOYING_32+" },
  "Encryption": { "PiiKeyId": "vault://kv/pii-key" }
}
```

--- Hujjat yakuni ---